

# Final Exam (Exceptional) Summer 2019\_2020

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1. The value of the limit

(3 Points)

$\lim_{x \rightarrow 0^+} \frac{\tan^{-1} \sqrt{x}}{\ln(1 + \sqrt{x})}$  is

- ☐ -1
- ☒ 1 ✓
- ☐ 2
- ☐ -2

2. The indefinite integral

(3 Points)

$\int \frac{\sqrt{x-2}}{x+2} dx$  equals to

- ☒  $2\sqrt{x-2} - 4 \tan^{-1} \frac{\sqrt{x-2}}{2} + C$  ✓
- ☐  $2\sqrt{x-2} - 8 \tan^{-1} \frac{\sqrt{x-2}}{2} + C$
- ☐  $2\sqrt{x-2} + 4 \tan^{-1} \frac{\sqrt{x-2}}{2} + C$
- ☐  $2\sqrt{x-2} - 4 \tan^{-1} \sqrt{x-2} + C$

3. The improper integral

(3 Points)

$\int_1^{\infty} \frac{1}{x(1 + (\ln x)^2)} dx$

- ☒ converges to  $\frac{\pi}{2}$  ✓
- ☐ converges to  $\frac{\pi-1}{2}$
- ☐ converges to  $\frac{\pi}{4}$
- ☐ diverges

4. Using the substitution

(3 Points)

$t = \tan \frac{x}{2}$  the definite integral  $\int_{\pi/3}^{\pi/2} \frac{1}{1 + \cos x} dx$  transforms to

- ☐  $\int_{1/\sqrt{3}}^1 \frac{dt}{t^2}$
- ☐  $\int_{1/\sqrt{3}}^1 \frac{dt}{1 - t^2}$
- ☐  $\int_{\sqrt{3}}^1 \frac{dt}{t^2}$
- ☒  $\int_{1/\sqrt{3}}^1 dt$  ✓

5. The sequence

(3 Points)

$a_n = \sqrt[4]{2^n + 3^n + 1^n}$

- ☐ converges to 4
- ☐ converges to 2
- ☒ converges to 3 ✓
- ☐ diverges

6. For what vales of c is the series

(3 Points)

$\sum \sqrt[n]{n^c}$  convergent?

- ☒  $c < -3$  ✓
- ☐  $c > -3$
- ☐  $c > 3$
- ☐  $c < 3$

7. Which of the following claims is NOT true for a series

(4 Points)

$\Sigma a_n$  with positive terms whose sequence of the partial sums  $s_n$  is bounded above?

- ☐  $\Sigma a_n$  is convergent
- ☒  $\Sigma a_n$  is divergent ✓
- ☐  $\lim_{n \rightarrow \infty} a_n = 0$
- ☐  $s_n$  is monotonic

8. The definite integral to calculate the volume of the solid obtained by rotating the region

(4 Points)

bounded by  $y = e^x$ ,  $x = 0$  and  $y = e$  about the line  $y = -1$  is

- ☒  $\pi \int_0^1 ((e+1)^2 - (e^x+1)^2) dx$  ✓
- ☐  $\pi \int_0^1 ((e+1)^2 - (e^x+1)^2) dx$
- ☐  $\pi \int_0^1 ((e^x+1)^2 + 1) dx$
- ☐  $\pi \int_0^1 (4 - (e^x+1)^2) dx$

9. Select the integral that represents the length of the curve

(3 Points)

$y = \tan x$  from  $(0,0)$  to  $(\frac{\pi}{4}, 1)$

- ☒  $\int_0^{\pi/4} \sqrt{1 + \sec^4 x} dx$  ✓
- ☐  $\int_0^{\pi/4} \sqrt{1 + \sec^2 x} dx$
- ☐  $\int_0^1 \sqrt{1 + \sec^4 x} dx$
- ☐  $\int_0^1 \sqrt{1 + \sec^2 x} dx$

10. Find all values of the parameter t, such that the parametric curve

(3 Points)

$x = e^t(t+1)$ ,  $y = t^3 + 3t^2$  has horizontal tangent line at the point  $(x(t), y(t))$ .

- ☐  $t = 0$ ;  $t = -2$
- ☐  $t = -2$
- ☒  $t = 0$  ✓
- ☐  $t = 2$

11. The surface area of the solid obtained by rotating the curve

(3 Points)

$y = \cosh x$ ,  $0 \leq x \leq 1$  about the  $x$ -axis is equal to

- ☐  $2\pi \int_0^1 x \sqrt{1 + \sinh^2 x} dx$
- ☒  $2\pi \int_0^1 \cosh x \sqrt{1 + \sinh^2 x} dx$  ✓
- ☐  $2\pi \int_1^{\cosh 1} \cosh x \sqrt{1 + \sinh^2 x} dx$
- ☐  $2\pi \int_0^1 x \sqrt{\cosh^2 x + \sinh^2 x} dx$

12. The equation of the tangent line at the point (1,4) to the parametric curve given by

(3 Points)

$x = \frac{1}{2}t^2 - t + 1$ ,  $y = t^2$  is

- ☒  $y = 4x$  ✓
- ☐  $y = -4x + 8$
- ☐  $y = 4x - 8$
- ☐  $y = -4x - 2$

13. The sum of the series

(3 Points)

$\Sigma_{n=0}^{\infty} (-1)^n \frac{(\ln 2)^n}{n!}$  is equal to

- ☒  $\frac{1}{2}$  ✓
- ☐  $-\frac{1}{2}$
- ☐  $\sqrt{e}$
- ☐  $\frac{1}{\sqrt{e}}$

14. The interval of convergence of the power series

(4 Points)

$\Sigma_{n=1}^{\infty} \frac{(2x+1)^n}{5^n \sqrt{n}}$  is

- ☐  $(-3, 2)$
- ☒  $[-3, 2)$  ✓
- ☐  $(-3, 2]$
- ☐  $[-3, 2]$

15. Which of the following series is conditionally convergent?

(3 Points)

- ☒  $\Sigma \frac{(-1)^n}{\sqrt[n]{n}}$  ✓
- ☐  $\Sigma \frac{\cos n}{n\sqrt{n}}$
- ☐  $\Sigma \frac{(-1)^n \tan^{-1} n}{n^2 + 1}$
- ☐  $\Sigma (\frac{9}{8})^n$

16. The Maclaurin series for the function

(3 Points)

$\ln(3+x)$  for  $|x| < 3$  is

- ☒  $\ln 3 + \Sigma_{n=1}^{\infty} (-1)^{n-1} \frac{x^n}{n3^n}$ , ✓
- ☐  $\ln 3 + \Sigma_{n=1}^{\infty} (-1)^n \frac{x^n}{3^n}$
- ☐  $\ln 3 + \Sigma_{n=1}^{\infty} (-1)^n \frac{x^n}{n3^n}$
- ☐  $\ln 3 + \Sigma_{n=2}^{\infty} (-1)^{n-1} \frac{x^n}{(n-1)3^n}$ ,

17. The Cartesian equation of the polar curve given by

(3 Points)

$r = \tan \theta \sec \theta$  is

- ☒  $y = x^2$  ✓
- ☐  $x = y^2$
- ☐  $x^2 = y^2$
- ☐  $xy = 1$

18. The area of the region that lies inside both polar curves

(4 Points)

$r = 1 - \cos \theta$  and  $r = \cos \theta$  is equal to

- ☒  $\int_0^{\pi/3} (1 - \cos \theta)^2 d\theta + \int_{\pi/3}^{\pi/2} \cos^2 \theta d\theta$  ✓
- ☐  $\int_0^{\pi/3} ((1 - \cos \theta)^2 + \cos^2 \theta) d\theta$
- ☐  $\int_{\pi/3}^{\pi/2} ((1 - \cos \theta)^2 + \cos^2 \theta) d\theta$
- ☐  $\int_0^{\pi/6} (1 - \cos \theta)^2 d\theta + \int_{\pi/6}^{\pi/2} \cos^2 \theta d\theta$

19. The y-coordinate of the centroid of the region bounded by

(4 Points)

$y = x^2 - 1$  and  $y = 0$  is

- ☐  $\frac{2}{5}$
- ☐ 0
- ☒  $-\frac{2}{5}$  ✓
- ☐  $\frac{16}{2\pi}$

20. The arc length of the curve

(4 Points)

$y = \frac{x^2}{2} - \ln(\sqrt[3]{x})$  for  $1 \leq x \leq a$  is

- ☒  $\frac{a^2}{2} + \frac{1}{4} \ln a - \frac{1}{2}$  ✓
- ☐  $\frac{a^2}{2} + \frac{1}{4} \ln a$
- ☐  $\frac{a^2}{2} + \ln a - \frac{1}{8} - \ln 2$
- ☐  $\frac{a^2}{2} + \ln a - \frac{1}{2} - e$

21. In the Taylor series about 2 for the function f(x) find the coefficient

(3 Points)

$c_6$  if  $f^{(6)}(2) = 54$  where  $f(x) = \Sigma_{n=0}^{\infty} c_n (x-2)^n$ .

- ☐  $\frac{1}{16}$
- ☐  $\frac{3}{8}$
- ☒  $\frac{3}{40}$  ✓
- ☐  $\frac{1}{8}$

22. How many terms of the series

(3 Points)

$\Sigma_{n=1}^{\infty} \frac{(-1)^n}{(2n+1)^3}$  should be added in order to find the sum with |error|  $< 10^{-3}$ ?

- ☐ 1
- ☐ 2
- ☐ 3
- ☒ 4 ✓

23. The value of the binomial coefficient

(3 Points)

$\binom{3/2}{3}$  is

- ☐  $-\frac{35}{16}$
- ☐  $-\frac{1}{2}$
- ☐  $-\frac{35}{8}$
- ☒  $-\frac{1}{16}$  ✓

24. The equation of the tangent line to the curve

(4 Points)

$r = \sin(2\theta)$  at  $\theta = \frac{\pi}{4}$  is

- ☒  $y = -x + \sqrt{2}$  ✓
- ☐  $y = 3x + \frac{2}{3}$
- ☐  $y = -2x + \frac{\sqrt{2}}{2}$
- ☐  $y = -3x + \frac{\sqrt{2}}{2}$

25. The length of the polar curve

(4 Points)

$r = 4 \cos \theta$ ,  $0 \leq \theta \leq \frac{\pi}{6}$  is

- ☐  $\pi$
- ☐  $\frac{4\pi}{3}$
- ☐  $\frac{\pi}{3}$
- ☒  $\frac{2\pi}{3}$  ✓

26. The length of the parametric curve given by the parametric equations

(4 Points)

$x = \cos t + \sin t$ ,  $y = \cos t - \sin t$ ,  $0 \leq t \leq 1$  is

- ☒  $\sqrt{2}$  ✓
- ☐  $2\sqrt{2}$
- ☐ 4
- ☐ 2

27. For what values of t is the parametric curve given by the parametric equations

(4 Points)

$x = t^2$ ,  $y = t^2 - 2t$  concave downward?

- ☒  $t \in (-\infty, 0)$  ✓
- ☐  $t \in (-\infty, \infty)$
- ☐  $t \in (0, \infty)$
- ☐  $t \in (-1, 1)$

28. The area bounded by the parametric curve

(3 Points)

$x = \cos^3 t$ ,  $y = \sin^3 t$ ,  $t \in [0, \frac{\pi}{2}]$  and the x-axis is given by

- ☐  $3 \int_0^{\pi/2} \sin^2 t \cos^4 t dt$
- ☐  $2\pi \int_0^{\pi/2} \sin^2 t \cos^4 t dt$
- ☒  $-3 \int_0^{\pi/2} \sin^4 t \cos^2 t dt$  ✓
- ☐  $6\pi \int_0^{\pi/2} \sin^2 t \cos^4 t dt$

29. The polar equation for the line y=5 is

(3 Points)

- ☐  $r = 5 \sec \theta$
- ☒  $r = 5 \csc \theta$  ✓
- ☐  $\theta = 5$
- ☐  $r = 5 \cos \theta$

30. The Maclaurin series for

(4 Points)

$\int \cos x^2 dx$  is

- ☒  $C + \Sigma_{n=0}^{\infty} \frac{(-1)^n x^{4n+1}}{(4n+1)(2n)!}$  ✓
- ☐  $C + \Sigma_{n=0}^{\infty} \frac{(-1)^n x^{4n}}{(4n+1)(2n)!}$
- ☐  $C + \Sigma_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)(2n)!}$
- ☐  $C + \Sigma_{n=0}^{\infty} \frac{(-1)^{n-1} x^{2n+3}}{(2n+3)(2n)!}$